Appendix S1 – Supplementary Information

Beaver engineering creates intense but bidirectional dissolved organic carbon hotspots in streams

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This document contains the material supporting the main findings of the manuscript (the hypothesised causal structure and tested hypotheses of the structural causal model, priors, residuals, posterior predictive checks, posterior parameter distributions, a stream-level classification of ΔDOC, the total effects of the drivers on ΔDOC, and an alternative SCM including lateral stream–wetland connectivity). The full description of data collection, variable derivation and model specification is given in the Methods of the main manuscript.

SCM: hypothesised causal structure

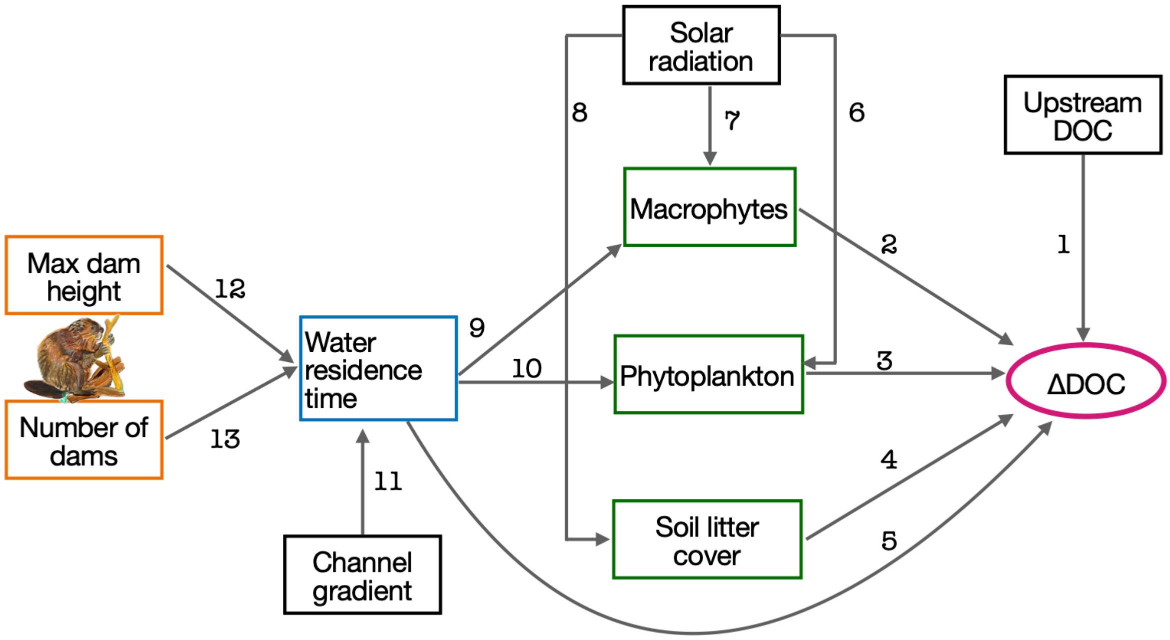

Figure S1: Directed acyclic graph (DAG) of the structural causal model (SCM), with the hypothesised causal mechanisms numbered 1–13. Orange boxes: beaver-engineering variables; the blue box: stream hydrology; green boxes: primary producer variables; black boxes: parent drivers not influenced by other variables; the magenta ellipse: the outcome ΔDOC. Each numbered edge corresponds to one tested hypothesis in Table S1. Beaver illustration credits to © Sarah Edmonds Illustration.

SCM: tested hypothesis

Table S1: Tested hypotheses for the structural causal model (SCM). The causal-path numbers correspond to the numbered edges of the DAG in Figure S1 (and Figure 1b of the main document).

Causal path	**Predictor**	**Response**	**Hypothesis**
1	DOC upstream	ΔDOC	If DOC upstream increases, then ΔDOC decreases because beaver ponds can act as sink, retaining or transforming DOC through processes such as microbial degradation, sedimentation, and photodegradation, thereby reducing downstream DOC concentrations relative to upstream inputs (Larson et al. 2007; Wohl 2013; Johnston 2014).
2	Macrophytes	ΔDOC	If macrophyte abundance increases, we expect ΔDOC to increase because the decomposition of macrophyte biomass releases labile dissolved organic carbon, enhancing downstream concentrations (Reitsema et al. 2018).

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Table S1 (continued)

Causal path	Predictor	Response	Hypothesis
3	Phytoplankton	ΔDOC	If phytoplankton abundance increases, then ΔDOC increases is likely to increase, because phytoplankton contribute to DOC content through exudation during growth and release upon cell lysis, thereby enhancing downstream DOC levels relative to upstream inputs (Søndergaard et al. 1985; Sell and Overbeck 1992).
4	Soil litter cover	ΔDOC	If soil litter cover increases, then ΔDOC increases as decomposition of litter releases dissolved organic compounds into the water (Aitkenhead et al. 1999; Cincotta et al. 2019).
5	Water residence time	ΔDOC	If water residence time increases, then ΔDOC increases, because longer water residence times allow for more extensive microbial decomposition and processing of dissolved organic carbon, enhancing downstream DOC concentrations (Jepsen et al. 2019; Abbasi et al. 2024).
6	Solar radiation	Phytoplankton	If solar radiation increases then phytoplankton abundance is expected to increase because greater light availability enhances photosynthetic rates (Reynolds 2006).
7	Solar radiation	Macrophytes	If solar radiation increases, then macrophyte abundance increases because higher light availability enhances photosynthetic activity and supports plant growth (Feuchtmayr et al. 2009).
8	Solar radiation	Litter cover	If solar radiation increases, as in summer, then soil litter cover decreases because lower rates of tree leaf fall coincide with enhanced decomposition driven by higher temperatures and solar radiation (Gessner et al. 1999).
9	Water residence time	Macrophytes	If water residence time increases, macrophyte abundance may increase because longer residence time allows for higher nutrient retention, promoting plant growth and increasing the likelihood of macrophyte establishment and persistence (Manolaki et al. 2021).
10	Water residence time	Phytoplankton	If water residence time increases, phytoplankton abundance may increase because longer residence time allows for greater nutrient retention and stabilization, which can support enhanced phytoplankton growth, particularly in systems with nutrient limitation (Finger et al. 2007).
11	Channel gradient (terrain slope)	Water residence time	If valley slope increases, then water residence time decrease because steeper terrain causes water to move more quickly through the landscape, reducing the time available for subsurface and surface interactions (McGuire et al. 2005).

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Table S1 (continued)

Causal path	**Predictor**	**Response**	**Hypothesis**
12	Max dam height	Water residence time	If dam height increases, water residence time increases because taller dams impound more water, expanding ponded areas and slowing streamflow, which prolongs the time water remains within the stream system (Majerova et al. 2015; Dittbrenner et al. 2022).
13	Number of dams	Water residence time	If the number of beaver dams increases, water residence time is expected to increase because dams slow down streamflow and create ponded areas that retain water for longer periods (Westbrook et al. 2006; Nyssen et al. 2011; Majerova et al. 2015).

Structural Causal Model: the priors

To ensure the model was not over-constrained by prior knowledge, we used weakly informative priors for all parameters. This approach allows the data to inform the model without assuming strong beliefs about the parameter values beforehand (Figure S2). We note, however, that for macrophytes, phytoplankton, and litter cover—variables measured in 16 of the 176 streams—we used more informative priors that incorporate established ecological knowledge about their influence on DOC concentrations (Figure S2 panel b).

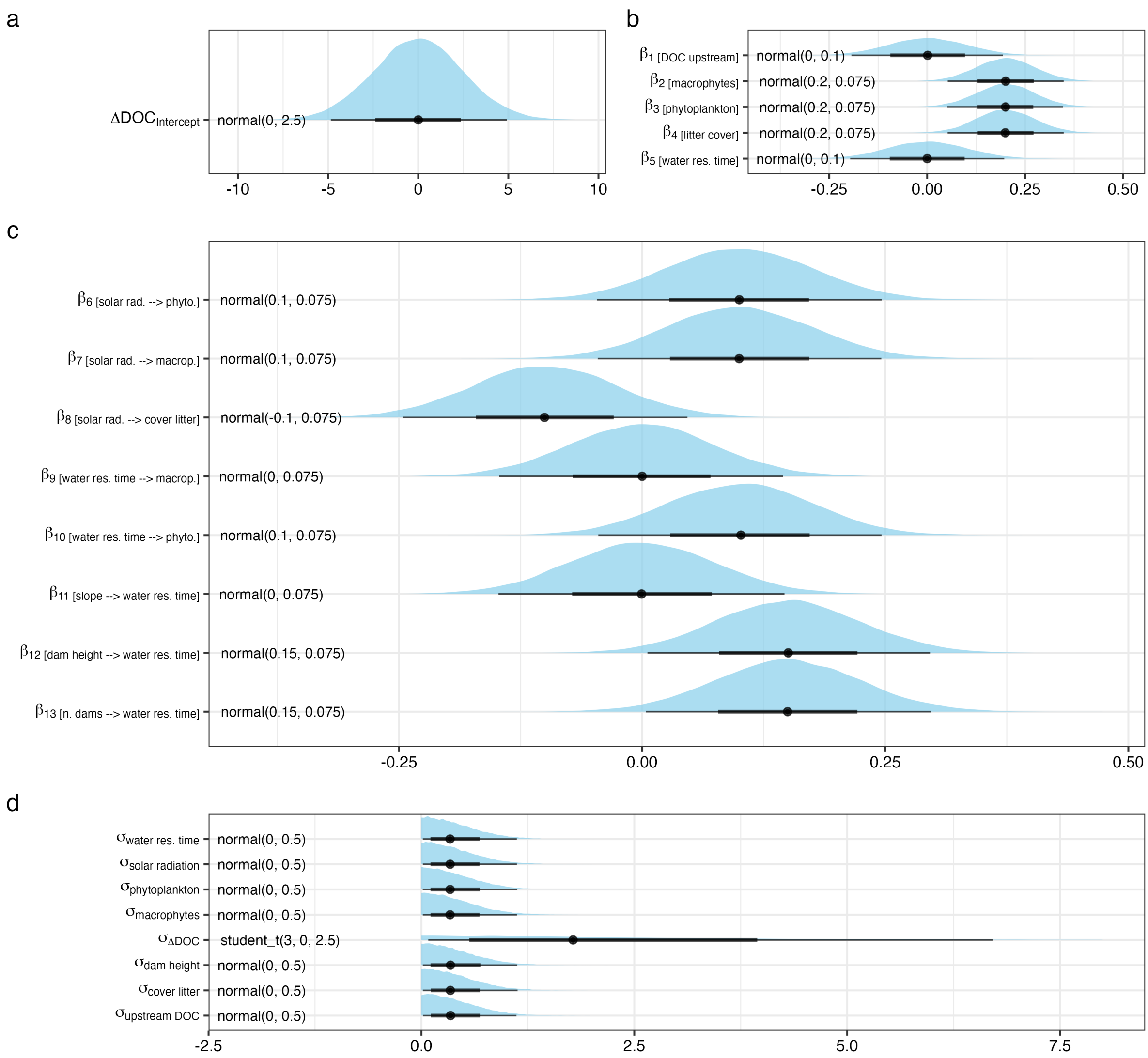

Figure S2: Half-eye plots display the prior distributions used in the model. The shaded areas (slabs) represent the probability density of each prior distribution, illustrating their shape and spread. The central dot indicates the median of the distribution, while the thick and thin horizontal lines represent the 66% and 95% credible intervals, respectively. Text labels to the left of each distribution indicate the corresponding prior specification.

Structural Causal Model: the residuals

We can evaluate the model fit by looking at the difference between observed data and their predictions. That are the residuals (r_i).

$$r_i = y_i - y_i^{pred} \tag{S1}$$

If the model is a reasonable representation of the observational process that originates the data then the residuals should look roughly randomly scattered in comparison to a horizontal line.

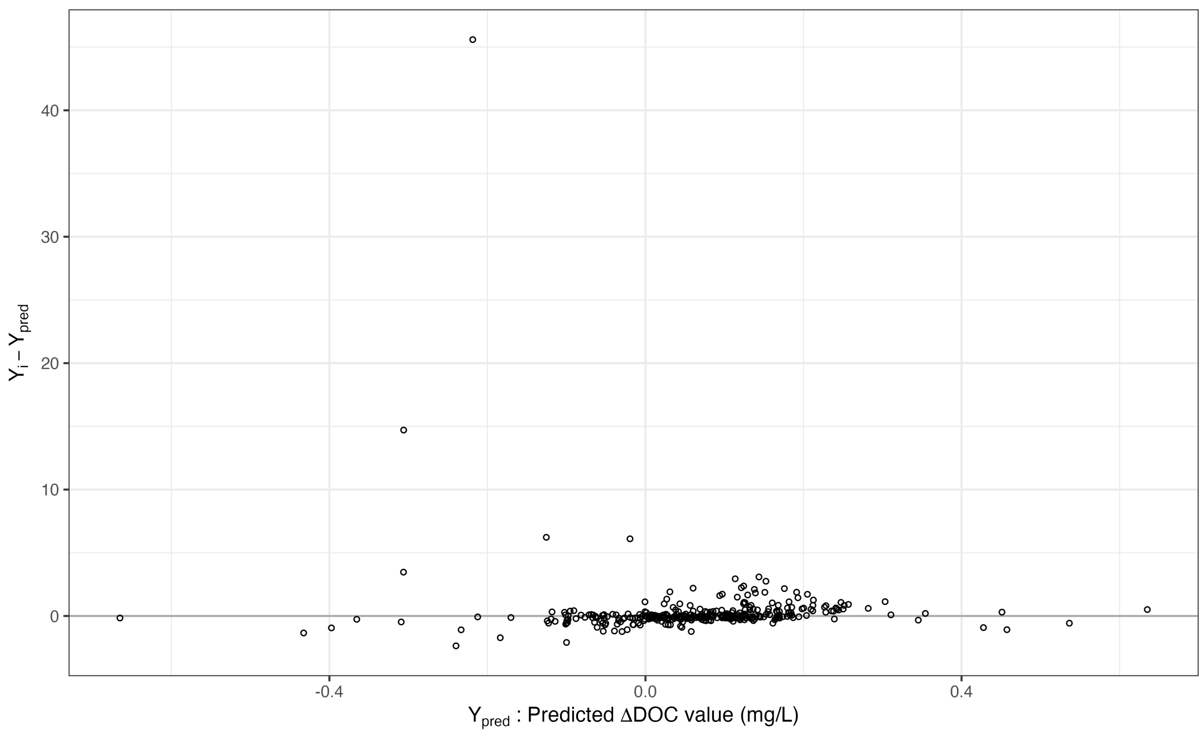

Figure S3: Residual plot showing the relationship between model-predicted ΔDOC values ($mg\,L^{-1}$) and corresponding residuals. Points represent the difference between observed and predicted values; a random scatter around zero (grey line) indicates good model fit and no systematic bias in predictions.

Structural Causal Model: posterior predictive checks

Posterior predictive checks assess how well the fitted SCM model reproduces observed data by comparing the observed values to data simulated from the model's posterior predictive distribution.

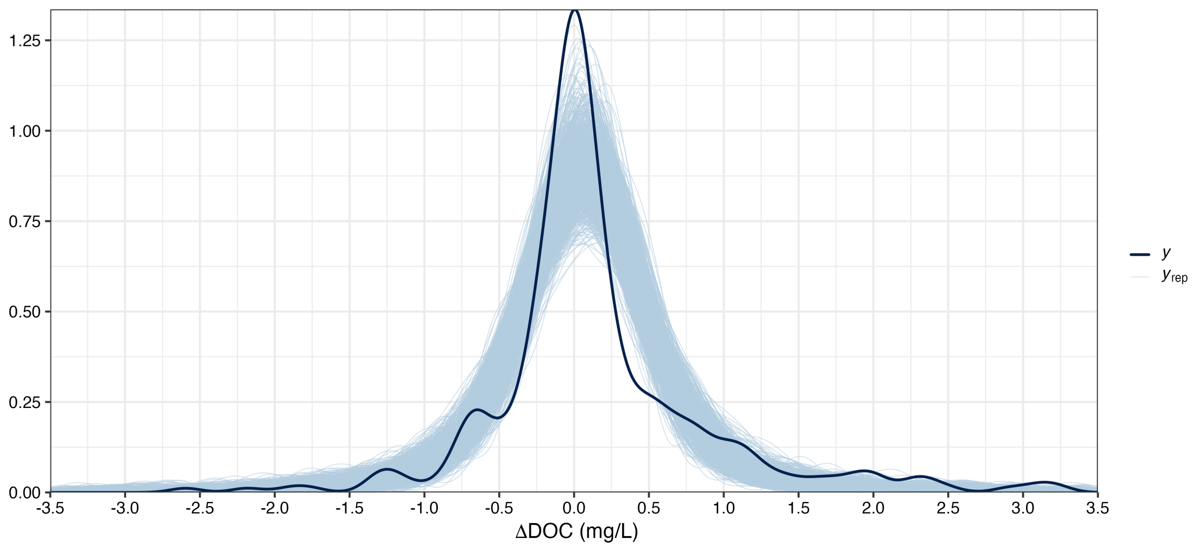

Figure S4: Posterior predictive checks of the mean and standard deviation statistics for the response variable delta DOC, with 1000 posterior draws used to generate the predictive distribution. This plot provides a visual assessment of the model's fit by comparing the observed data (dark blue) against the model's predictions (light blue) from the posterior distribution.

Structural Causal Model: posterior parameters distribution

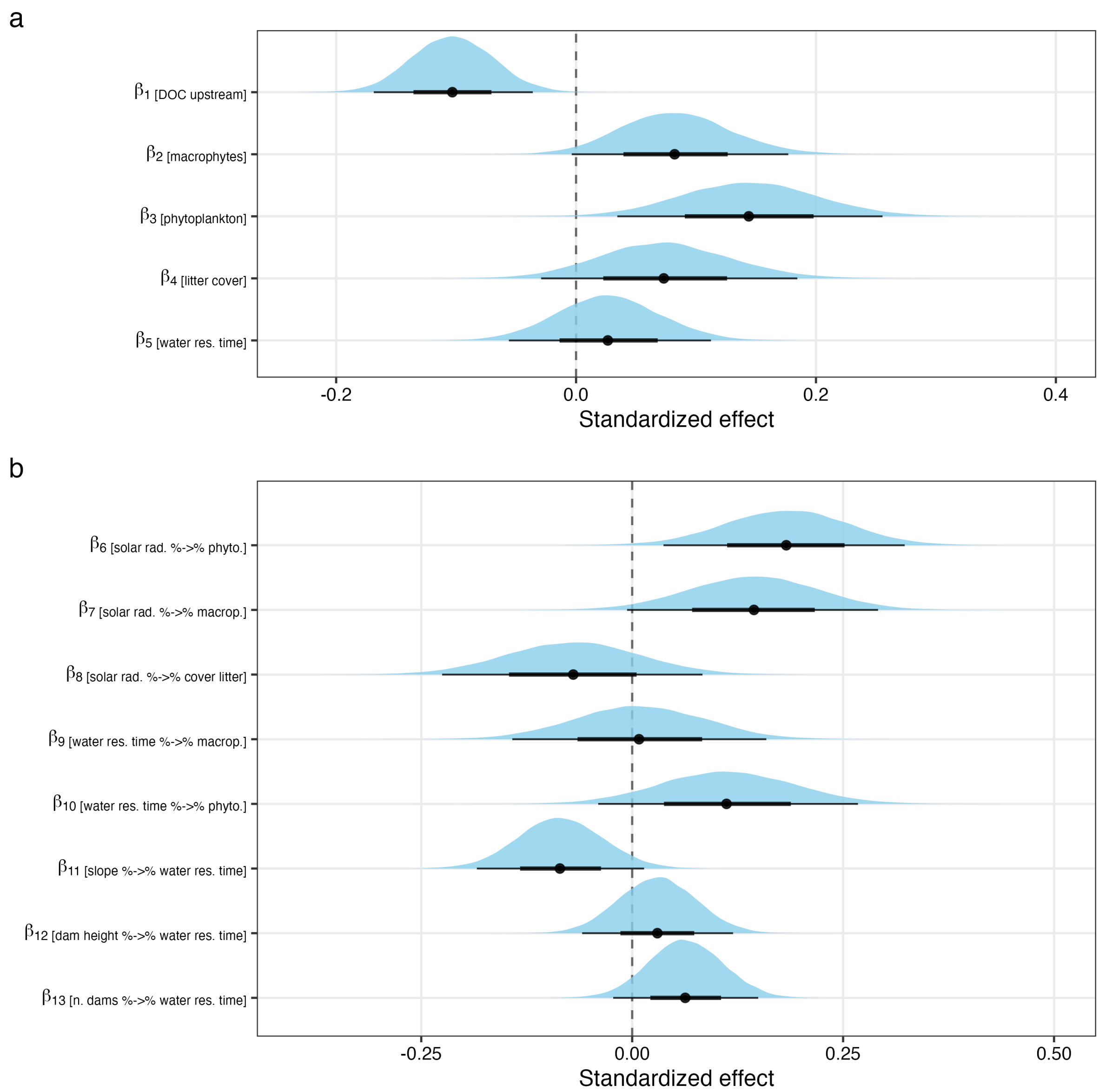

Figure S5: Posterior parameters distributions of the fitted SCM. Median shown with point; 66% and 95% credible intervals shown with black bars. Parameters that have more than 90% probability mass on either side of zero are considered to have a positive or negative influence on the response variable. a) The direct effects on ΔDOC b) The indirect effects on ΔDOC.

Structural Causal Model: stream-level classification of ΔDOC

To avoid conflating the posterior predictive probability of a response (Figure 1a, main document) with the proportion of streams showing that response, we classified each stream individually using the structural causal model (SCM), which includes a stream-level random intercept. (As a consistency check, the posterior predictive distribution of ΔDOC from the SCM for a typical stream – predictors at their mean, stream effects integrated out – gives 12% sink, 76% negligible and 12% source responses, in close agreement with the season-specific measurement-error model of Figure 1a.) For each of the 179 streams we computed the posterior distribution of the expected ΔDOC (population-level effects plus the stream-level intercept, averaged over the stream's observations; 2,000 posterior draws) and assigned the stream to the region (sink < -0.5 mg L^{-1}, ROPE ± 0.5 mg L^{-1}, source > 0.5 mg L^{-1}) that held the majority of its posterior mass.

The expected ΔDOC of all 179 streams fell within the ROPE (posterior medians from −0.40 to 0.38 mg L^{-1}; median width of the 90% credible interval 0.41 mg L^{-1}; Figure S6), reflecting the small average effect (intercept 0.00 mg L^{-1}; 95% BCI −0.12; 0.12) and the small between-stream standard deviation (0.05 mg L^{-1}; 95% BCI 0.00; 0.14). The heavy-tailed posterior predictive distribution of Figure 1a thus captures the variability of individual samplings around a null average rather than consistent stream-level differences.

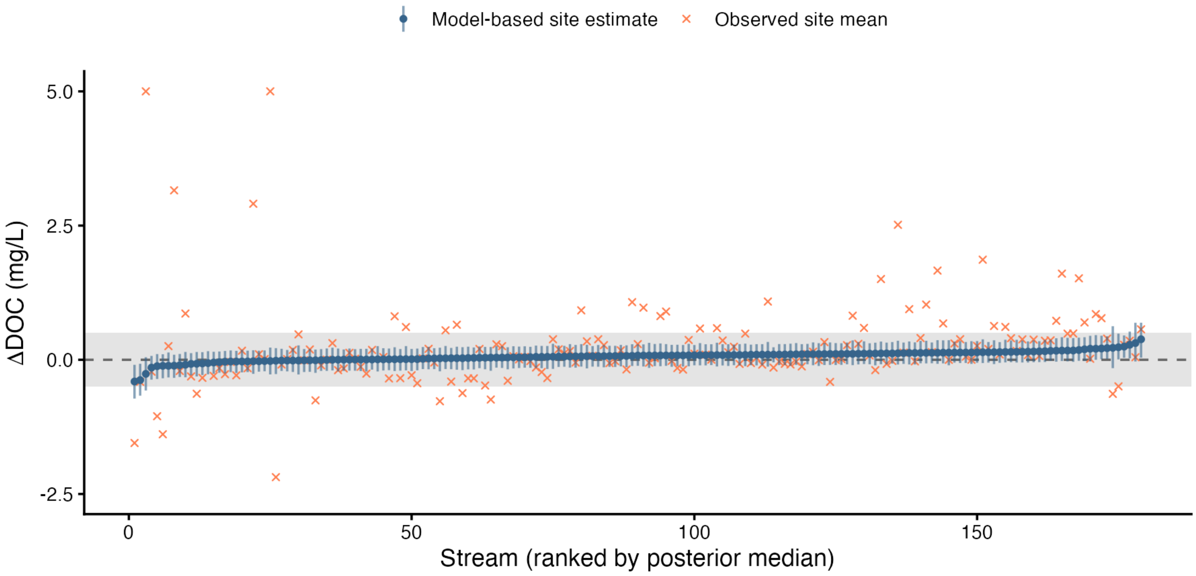

Figure S6: Stream-level ΔDOC. Blue points and bars: posterior median and 90% credible interval of the model-based expected ΔDOC of each stream (ranked by posterior median). Orange crosses: observed stream-averaged ΔDOC, shown for reference (one value of 29.9 mg L^{-1} is clipped at 5 mg L^{-1}). The grey band is the region of practical equivalence (±0.5 mg L^{-1}).

Structural Causal Model: total effects of the exogenous drivers on ΔDOC

The edges of the SCM (Figure 1b, main document) are direct, path-specific effects. To summarize the overall influence of each exogenous driver on ΔDOC we computed its *total effect* as the sum of the products of the coefficients along every open causal path from the driver to ΔDOC (e.g., for the number of dams: dams → WRT → ΔDOC, dams → WRT → macrophytes → ΔDOC, and dams → WRT → phytoplankton → ΔDOC), evaluated on each posterior draw (path tracing; Hayes 2018). All effects are on the standardized scale (change in ΔDOC, in standard deviations, per standard deviation of the driver).

Upstream DOC concentration had the largest total effect (−0.10; 90% BCI: −0.16; −0.05; $P(\beta < 0) > 0.99$), followed by solar radiation, which acts through all three primary-producer mediators (+0.04; 90% BCI: 0.01; 0.07; $P(\beta > 0) = 0.99$). The total effect of water residence time (+0.04; 90% BCI: −0.02; 0.11; $P(\beta > 0) = 0.87$) was positive but less certain. The total effects of the beaver-engineering variables, which act only through water residence time, were small: number of dams +0.003 (90% BCI: −0.002; 0.010; $P(\beta > 0) = 0.82$) and maximum dam height +0.001 (90% BCI: −0.003; 0.007; $P(\beta > 0) = 0.68$); channel gradient had a small negative total effect (−0.004; 90% BCI: −0.012; 0.002). Hence, the direction of the beaver-mediated DOC change is set primarily by the external context of the pond (incoming DOC load and light availability for primary producers), whereas the structural characteristics of the dam complex modulate it only weakly via water residence time (Figure S7).

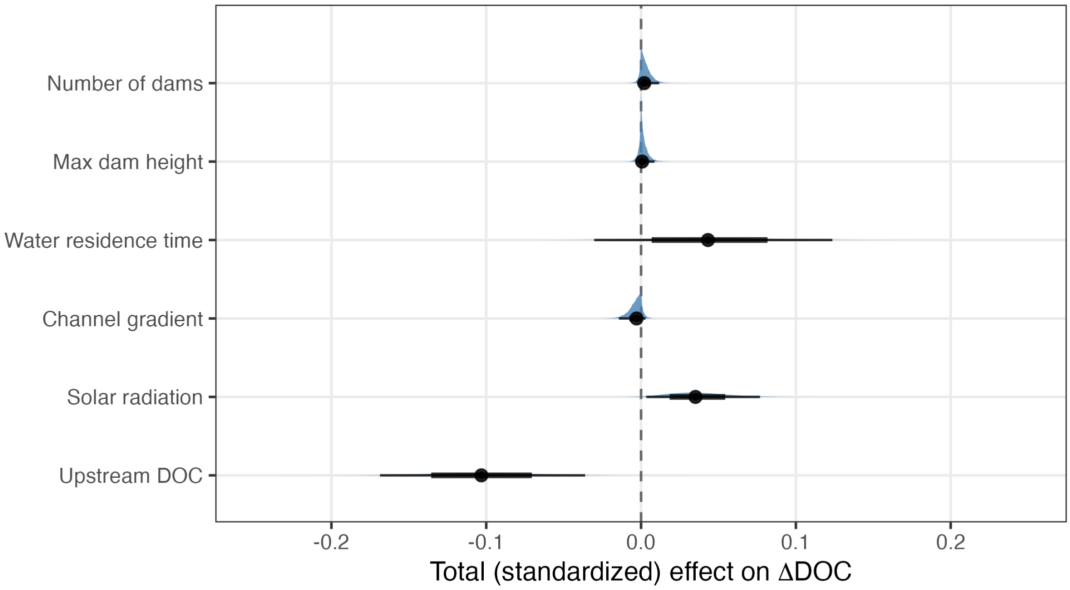

Figure S7: Posterior distributions of the total (standardized) effect of each exogenous driver on ΔDOC, obtained by tracing all open causal paths of the SCM. Points: posterior means; thick and thin bars: 66% and 95% credible intervals.

Alternative structural causal model: lateral stream–wetland connectivity

Because a DAG encodes a specific set of causal assumptions, we also tested an alternative, extended SCM that incorporates the lateral connectivity between the stream and adjacent wetlands as an additional driver of water residence time (WRT). Hydrologically connected off-channel storage zones (wetlands, floodplains) exchange water with the channel and are known to extend the residence time of water in the system (transient-storage theory; Bencala and Walters 1983). We therefore hypothesized that beaver-pond WRT varies with the degree of lateral stream–wetland connectivity and, consequently, that connectivity indirectly influences ΔDOC via WRT and primary producers.

Connectivity variable. For each site, we first determined whether a wetland was present within a 50 m buffer around the river reach between the upstream and downstream sampling locations, using the wetland layer (marshes, bogs, swamp forests, wet meadows, floodplains and fens) of the swissTLM3D topographic landscape model (SwissTopo 2024). For sites with a wetland, we then assessed visually on the swisstopo maps and orthophotos whether the wetland was hydrologically connected to the river water, or separated from it by roads, trails, tree lines or other barriers. Each site was thereby assigned to one of three categories: *No wetland* (no wetland within the buffer; $n = 136$ sites), *Wetland, not connected* (wetland present but not directly connected to the river; $n = 11$), and *Wetland, connected* (wetland present and directly connected to the river; $n = 13$). Twenty-two sites could not be classified and were excluded from this analysis (310 observations retained). Note that most of these wetlands are themselves strongly influenced by beaver damming and would likely not exist without beavers.

DAG and identification. In the extended DAG (Figure S8) lateral connectivity has no parents, hence there are no open back-door paths between connectivity and ΔDOC and the total effect of connectivity on ΔDOC is identified without adjustment (empty adjustment set, *dagitty*). The causal influence flows through two front-door paths: connectivity → WRT → macrophytes → ΔDOC and connectivity → WRT → phytoplankton → ΔDOC. Because the total effect of WRT on ΔDOC is likewise unconfounded, we fitted a parsimonious two-equation Bayesian model in *brms*: (i) ΔDOC (with the 0.283 mg L^{-1} measurement error, Student-t likelihood, site-level random intercept) regressed on WRT; (ii) WRT regressed on number of dams, maximum dam height, channel gradient, solar radiation and lateral connectivity (reference level: *No wetland*). Missing values in WRT and dam height were imputed within the model (`mi()`). Four chains were run (R-hat ≤ 1.005). The total effect of each connectivity level on ΔDOC was computed from the posterior draws as the product of the connectivity → WRT contrast and the WRT → ΔDOC coefficient.

Results. Relative to sites without wetland, the presence of an unconnected wetland increased WRT by 0.20 standardized units (90% BCI: −0.20; 0.59; $P(\beta > 0) = 0.80$) and a connected wetland by 0.01 (90% BCI: −0.35; 0.39). The WRT → ΔDOC path was positive but weak (0.09; 90% BCI: 0.01; 0.16). The resulting total effects of connectivity on ΔDOC were negligible: 0.02 (90% BCI: −0.02; 0.07) for unconnected and 0.00 (90% BCI: −0.04; 0.04) for connected wetlands (Figure S9). We therefore found no evidence that lateral stream–wetland connectivity modifies the beaver-mediated DOC change through the water-residence-time pathway. Given that only 7% and 6% of sites fell into the connected and unconnected wetland categories, we regard this result as power-limited rather than as strong evidence of absence of an effect. Importantly, the estimates of the remaining paths were consistent with those of the main SCM (Figure 1b, main manuscript), indicating that our main conclusions are robust to this alternative causal structure.

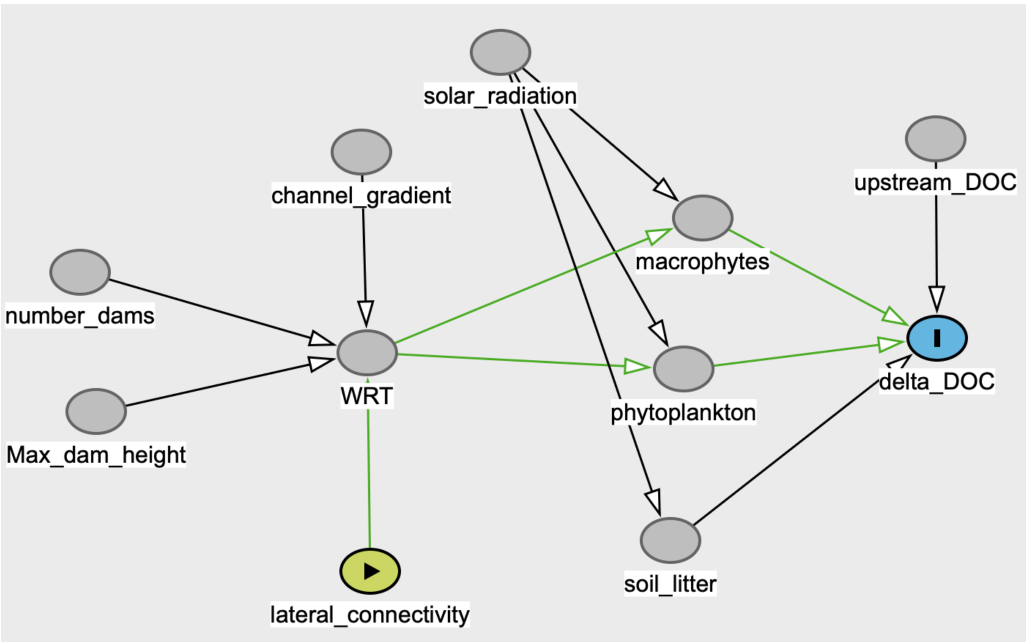

Figure S8: Extended directed acyclic graph (DAG) including lateral stream–wetland connectivity (exposure, green) as a driver of water residence time (WRT). Green edges mark the open front-door paths from connectivity to ΔDOC (outcome, blue).

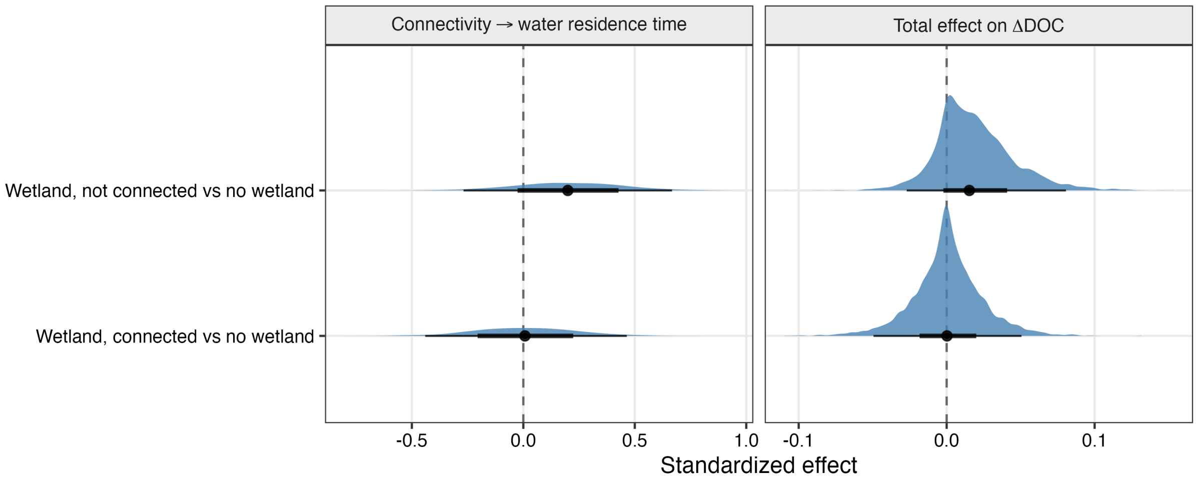

Figure S9: Posterior effects of lateral stream–wetland connectivity (relative to sites without wetland) on water residence time (left) and the total mediated effect on ΔDOC (right), controlling for beaver-dam variables, channel gradient and solar radiation. Points: posterior medians; thick and thin bars: 66% and 95% credible intervals. The dashed line marks zero.

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